

Calculus I Lesson 29B

1) Find $\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 - x}}$

a) $\frac{x}{\sqrt{x^2 - x}} = \frac{x / \sqrt{x^2}}{\sqrt{x^2 - x} / \sqrt{x^2}} = \frac{x / x}{\sqrt{x^2 / x^2 - x / x^2}} = \underline{\hspace{2cm}} ?$

Note: $\sqrt{x^2} = |x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$

Since $x \rightarrow \infty$, $\sqrt{x^2} = x$

b) $\lim_{x \rightarrow \infty} \left(\frac{x}{\sqrt{x^2 - x}} \right) = \underline{\hspace{2cm}} ?$

2) Find $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2 - x}}{5x - 1}$

a) $\frac{\sqrt{x^2 - x}}{5x - 1} = \frac{\sqrt{x^2 - x} / x}{5x/x - 1/x} = \frac{\sqrt{x^2 - x} / \sqrt{x^2}}{5 - 1/x} = \frac{\sqrt{x^2/x^2 - x/x^2}}{5 - 1/x} = \underline{\hspace{2cm}} ?$

Note: $\sqrt{x^2} = |x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$

Since $x \rightarrow \infty$, $\sqrt{x^2} = x$

b) $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x^2 - x}}{5x - 1} \right) = \underline{\hspace{2cm}} ?$

3) Find $\lim_{x \rightarrow \infty} \frac{4x^2 - x}{\sqrt{x} - 1}$

a) $\frac{4x^2 - x}{\sqrt{x} - 1} = \frac{4x^2 / \sqrt{x} - x / \sqrt{x}}{\sqrt{x} - 1 / \sqrt{x}} = \frac{4x^2 / x^{1/2} - x / x^{1/2}}{\sqrt{x / x} - 1 / x} = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow \infty} \left(\frac{4x^2 - x}{\sqrt{x} - 1} \right) = \underline{\hspace{2cm}} ?$

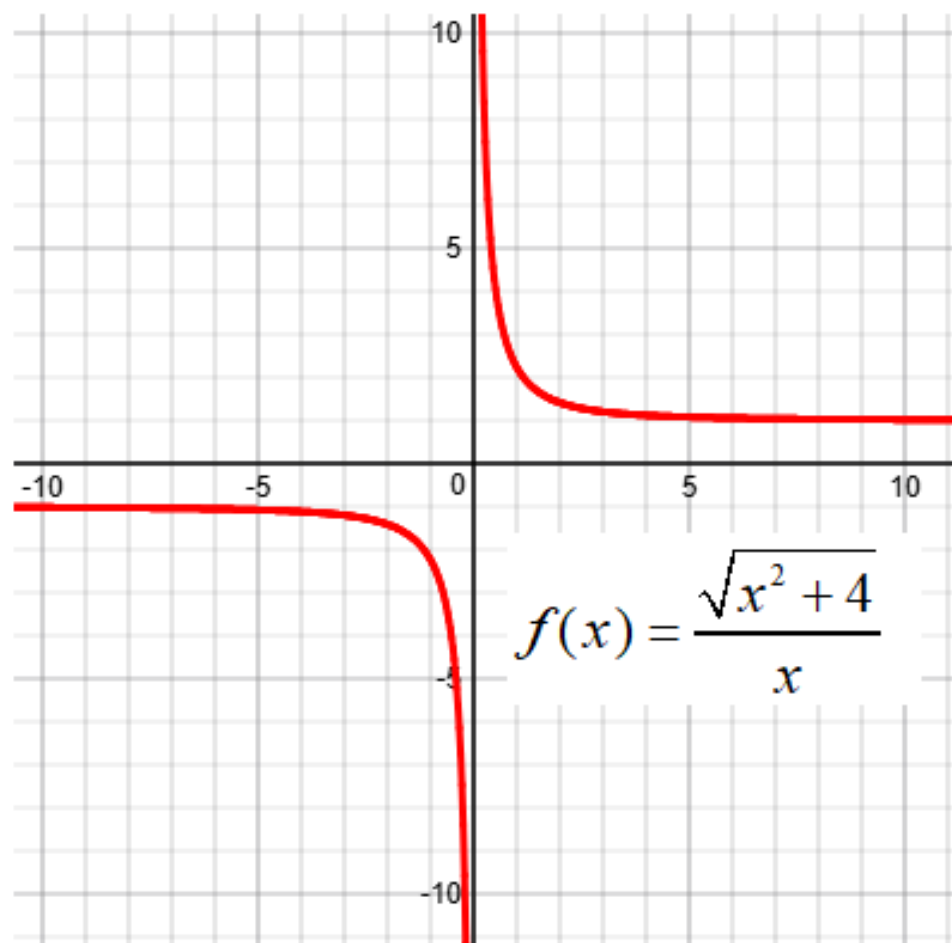
4) Find $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 4}}{x}$

Note: $\sqrt{x^2} = |x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$

Since $x \rightarrow -\infty$, $\sqrt{x^2} = -x$

a) $\frac{\sqrt{x^2 + 4}}{x} = \frac{\sqrt{x^2 + 4} / (-x)}{x / (-x)} = \frac{\sqrt{x^2 + 4} / \sqrt{x^2}}{-1} = \frac{\sqrt{x^2 / x^2 + 4 / x^2}}{-1} = \underline{\quad ? \quad}$

b) $\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{x^2 + 4}}{x} \right) = \underline{\quad ? \quad}$



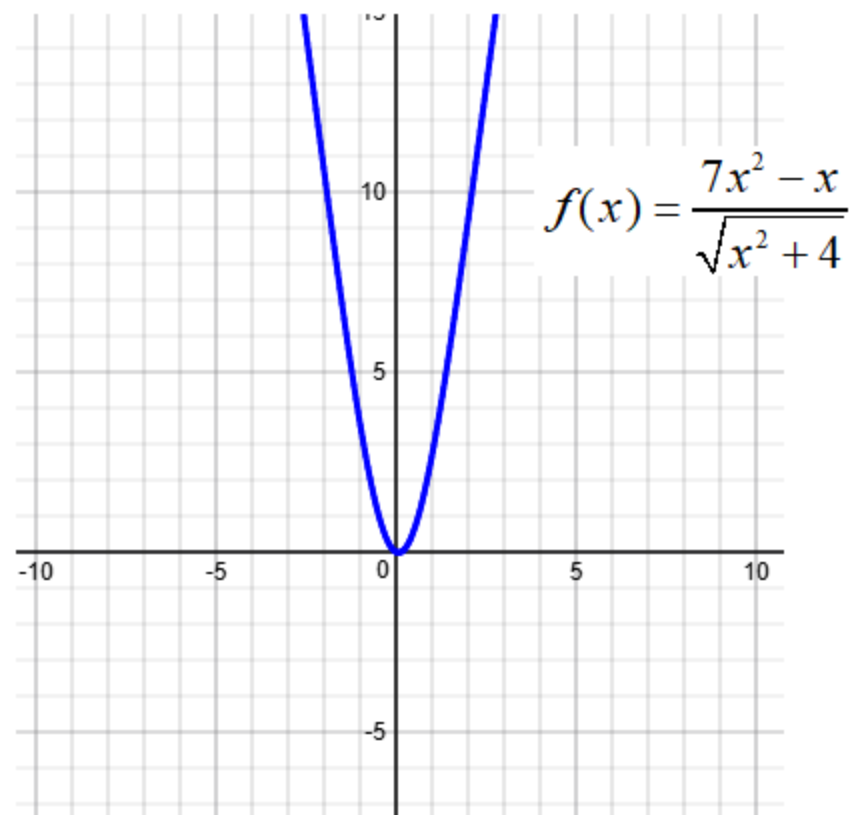
5) Find $\lim_{x \rightarrow -\infty} \frac{7x^2 - x}{\sqrt{x^2 + 4}}$

Note: $\sqrt{x^2} = |x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \end{cases}$

Since $x \rightarrow -\infty$, $\sqrt{x^2} = -x$

a) $\frac{7x^2 - x}{\sqrt{x^2 + 4}} = \frac{7x^2 / \sqrt{x^2} - x / \sqrt{x^2}}{\sqrt{x^2 + 4} / \sqrt{x^2}} = \frac{7x^2 / (-x) - x / (-x)}{\sqrt{x^2 / x^2 + 4 / x^2}} = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow -\infty} \left(\frac{7x^2 - x}{\sqrt{x^2 + 4}} \right) = \underline{\hspace{2cm}} ?$



6) Find $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^4 + 2}}{x^3 - 2}$

a) $\frac{\sqrt{x^4 + 2}}{x^3 - 2} = \frac{\sqrt{x^4 + 2} / x^3}{x^3 / x^3 - 2 / x^3} = \frac{\sqrt{x^4 + 2} / \sqrt{x^6}}{1 - 2 / x^3} = \frac{\sqrt{x^4 / x^6 + 2 / x^6}}{1 - 2 / x^3} = \underline{\hspace{2cm}} ?$

b) $\lim_{x \rightarrow -\infty} \left(\frac{\sqrt{x^4 + 2}}{x^3 - 2} \right) = \underline{\hspace{2cm}} ?$

